

ALEXANDER GILBREATH

MACHINE LEARNING • AI • SOFTWARE ENGINEERING

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Junior AI student at UC San Diego who builds, trains, and validates ML systems end to end — a GPT-style transformer from scratch in PyTorch, GPU-trained CNNs, and a shipped AI-vision iOS app — on a C/C++ systems foundation. U.S. work authorized; no sponsorship required.

EDUCATION & CERTIFICATIONS

University of California, San Diego — B.S. Artificial Intelligence *Junior standing* Expected June 2028 • GPA 3.6
Coursework: Intro to AI • Systems Prog. in C (CSE 29) • Computer Org. & Assembly (CSE 30, A) • Discrete Math (CSE 20, A) • Data Structures (CSE 12)
In progress: Mathematics for Algorithms & Systems (CSE 21) • Optimization Methods for Data Science (MATH 173A)
Google AI Professional Certificate Google / Coursera, 2026 • **AWS Certified AI Practitioner (AIF-C01)** *in progress, 2026*

TECHNICAL SKILLS

Languages	Python, C, C++, Java, MATLAB, TypeScript, ARM assembly
ML & AI	PyTorch, deep learning, computer vision, CNNs, transformers & attention, hyperparameter tuning, model evaluation, LLM APIs
Systems & GPU	CUDA GPU training, Linux, C/C++ systems programming, memory management, embedded firmware, performance tuning
Tools	Git, GitHub, Jupyter, gcc, gdb, valgrind, Make, Xcode, Cursor, React, Vite
Testing	Unit testing, memory-leak checking, ground-truth dataset construction, accuracy benchmarking (F1, precision/recall)

PROJECTS

GPT Language Model from Scratch *PyTorch, Python, CUDA* 2026

- Built a 30M-parameter decoder-only transformer with no high-level model libraries: multi-head attention, tokenization, full training loop.
- Cut training loss 37% (6.11 → 3.87) through hyperparameter sweeps over learning rate, batch size, and context length on GPU.

Disaster-Damage Classification *ACM AI • PyTorch, CNNs, Computer Vision* 2025

- Raised accuracy on satellite damage imagery on a 3-person team by designing a dense CNN (4 conv → 2 FC layers) and tuning it on GPU.

Acne Lesion Classifier *PyTorch, CNNs, Medical Imaging* 2025

- Reached 0.89–0.94 per-class F1 across six lesion classes, validated via confusion matrices and per-class precision/recall.

CompGym — AI Nutrition & Training Log *iOS, AI Vision* 2026

- Helped ship an iOS app that identifies food and estimates macros from a single photo, contributing from requirements through release testing.
- Improved identification and portion accuracy by hand-measuring ground-truth nutrition datasets and benchmarking predictions against them.

Posture-Correction Wearable *C++, Embedded, Hardware-Software Integration* 2026

- Built a real-time posture-alert wearable: C++ firmware computes spinal angle from a 3-axis accelerometer against calibrated thresholds.

AI Accessibility Research *UC San Diego, advisor Sina Radi* 2025 – Present

- Contribute to applied CV research building AI communication tools for a nonverbal user with severe autism, through model validation.

Systems & Software Engineering *C, Linux, CSE 29 / 30* 2026

- Delivered a leak-free malloc/free library in C (best-fit allocation, 16-byte alignment, public `vmLib.h` API), verified via valgrind and unit tests.
- Built a Unix shell (fork/execvp/wait, persistent history) and a linked-list calendar with conflict-detection invariants, both assert-tested.

EXPERIENCE

A1Base *Y Combinator-backed AI startup • AI Product & Campus Ambassador* 2026 – Present

- Work with the team building “Sunny,” an AI campus assistant that texts students contextual info on events, scheduling, and campus life.
- Lead UCSD user testing and deliver weekly feedback on model accuracy and gaps, while driving grassroots growth of the user base.

UC San Diego Environmental Health & Safety *Course Auditor* 2025 – Present

- Audit AI-generated training content for errors and hallucinations before campus-wide release; train Resident Assistants on fixes.

RedCoin Capital *Research Intern* — translated research on subnet architectures and decentralized-AI infrastructure into technical briefs. 2025

Tritons Thrift *Founder* — iOS App & Student Resource 2025 – Present

- Building a native iOS app in Xcode — a campus resale marketplace offered as an official UCSD Associated Students resource.
- Grew a resale operation (~\$5,000 in five months) into a tech product now scaling into an official AS partnership.

CAMPUS LEADERSHIP & CLUBS

Engineering Innovators & Entrepreneurs Club (EIEC) *President* 2025 – Present

- Elected Financial Chair, Co-President (as a freshman), then President; lead hackathons and a venture cohort with the Qualcomm Institute.
- Co-presented the 1st Physician AI Hackathon with the Society of Physician Entrepreneurs; covered by PR Newswire and SD Business Journal.
- Founded and solo-organized the club's own hackathon — 40 builders, \$1,000 prize pool — securing sponsors and judges.

Triton Engineering Student Council (TESC) *Marketing Intern* May 2026 – Present

- Market events for ~5,300 students across 18 organizations (DECaf career fair, HackXX); helped bring Adobe and Google on as Fall 2026 partners.

Triton Council for Entrepreneurs *Founder & Principal Member* — shape direction and programming for UCSD student founders. Current

AI Collective (AIC) *Financial Analyst & Media Coordinator* — run media and budgets for hackathons and project nights. Mar 2026 – Present

AWARDS & REFERENCES

Winner, 2025 OpenArt AI Hackathon *UC San Diego, presented by CSES* 2025

References: Prof. Adalbert Gerald Soosai Raj (CSE 29, CSE 8A) • Prof. Barry Brown (CSE 21) — full contact details available on request.